



Challenges and Solutions for Countering Emerging Security Threats to IoT Devices

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Article History:

Received: 16th Nov., 2024

Accepted: 18th Dec., 2024

Published: 30th Apr., 2025

Keywords:

IoT security, Encryption, Cyber threats, Data security, Authentication, Security solutions

How to cite this Article:

Naman Tiwari, Abhay Kumar Pandey and Abhilash Kokala (2025). Challenges and Solutions for Countering Emerging Security Threats to IoT Devices. *International Journal of Experimental Research and Review*, 47, 59-68.

DOI:

<https://doi.org/10.52756/ijerr.2025.v47.005>

Abstract: The Internet of Things (IoT) has transformed how devices communicate and interact, making daily life more connected and efficient. However, as IoT adoption increases, so do security risks. IoT devices are a target for cybercriminals who want to steal data, control operations, and disrupt services. This research examines the major security issues that IoT devices face and potential solutions to protect them from new threats. Insufficient authentication and authorization are one of the most significant obstacles in IoT security. Due to their lack of security updates and default passwords, many IoT devices are easy targets for cyberattacks. Additionally, the likelihood of data breaches is raised by the fact that IoT devices frequently transmit data without encryption. The lack of standard security protocols is another major problem because different manufacturers use different security measures, which creates inconsistencies and vulnerabilities. This paper discusses a variety of solutions to these issues, such as encryption methods, regular security updates, and strong authentication mechanisms. By ensuring that only authorized users can access IoT devices, using multi-factor authentication (MFA) can improve security. Attackers can't get their hands on sensitive data if it is encrypted. Additionally, real-time security threat detection and prevention can be made possible by utilizing AI and ML. Governments and organizations also play a vital role in improving IoT security. When designing and developing IoT devices, manufacturers may adhere to best practices if global security standards and regulations are established. Vulnerabilities can be further reduced by educating users about cybersecurity threats and best practices. In conclusion, a combination of robust authentication, encryption, regular updates, and global security policies is required to safeguard IoT devices from security threats. To create an IoT environment that is safer and more secure as technology advances, ongoing research and development are required.

Introduction

The Internet of Things (IoT) has changed how devices communicate with one another, making automation and seamless communication possible in a variety of industries. Wearable devices, industrial automation, and healthcare systems are just a few examples of the many ways in which the Internet of Things (IoT) has emerged as an essential component of contemporary society. The benefits of this development include increased convenience, improved decision-making, and increased

efficiency. However, the risk of cyber threats increases with IoT adoption (Alaba et al., 2017; Alwarafy et al., 2020). IoT devices' security flaws have become a major concern because cybercriminals frequently target them for their potential to be exploited for financial gain, data theft, or service disruption. As a result of their reliance on the Internet for communication and the rapid expansion of IoT networks, attackers now have new opportunities to break into systems, steal sensitive data, and disrupt essential services. IoT devices' inherent lack of robust



security mechanisms is one of the main reasons they are vulnerable to security threats (R et al., 2024). Many IoT devices lack the memory and processing power necessary to support robust encryption, authentication, and other security measures (Bharati and Podder, 2022; Khurana et al., 2024). Additionally, devices with weak or default passwords, software vulnerabilities that have not been patched, and inadequate security protocols are frequently deployed as a result of manufacturers prioritizing affordability and functionality over security. IoT ecosystems are frequently the target of malware attacks, distributed denial-of-service (DDoS) attacks, unauthorized access, and data breaches as a result. The risk is made worse by the interconnected nature of IoT devices because a single compromised device can be used as an entry point for hackers to break into an entire network. The absence of standardized security protocols and regulations is yet another significant obstacle to IoT security (Driss et al., 2021; Medishetty and K, 2023; Fernandez-Carames, 2024). IoT devices, in contrast to conventional computing systems, come from a wide variety of manufacturers and each employs its own security measures. Because of this fragmentation, it is challenging to establish uniform security standards. As a result, IoT devices take inconsistent approaches to authentication, data encryption, and firmware updates. It becomes difficult to ensure that all IoT devices meet a minimum level of cyber threat protection without a unified security strategy. In addition, Gubbi et al. (2013) found that many IoT devices are used in settings where security updates are either scarce or nonexistent. IoT devices frequently remain vulnerable to security risks as a result of manufacturers' reluctance or inability to provide timely updates, in contrast to smartphones and personal

computers, which receive software updates to patch vulnerabilities on a regular basis. IoT devices are a tempting target for cybercriminals due to the absence of ongoing security maintenance. IoT security is also associated with concerns about data privacy. Personal information, location details, and usage patterns are just a few examples of the vast amounts of data that IoT devices continuously collect, transmit, and process. The fact that this data is frequently stored on cloud servers raises the possibility of data breaches and unauthorized access. Cybercriminals have the ability to intercept and manipulate data if IoT devices lack robust encryption, which can lead to serious consequences like identity theft, financial fraud, and corporate espionage (Henze et al., 2016). Additionally, because many IoT applications rely on real-time monitoring and control, any security breach can have an immediate and potentially disastrous impact. In the healthcare industry, for instance, compromised IoT devices that are used in medical monitoring systems can lead to incorrect diagnoses, treatment errors, or even death. Unauthorized access to IoT-controlled machinery in an industrial setting can result in operational disruptions, equipment damage and safety risks (Jara et al., 2013; Jing et al., 2014). The potential consequences of IoT security breaches emphasize the significance of putting in place efficient security measures to safeguard these devices from emerging threats. It is essential to implement a combination of robust authentication mechanisms, encryption techniques, and proactive security measures in order to reduce the risks associated with IoT security threats. The use of multi-factor authentication (MFA) is one of the fundamental security practices for IoT devices, according to Rani et al. (2021). MFA significantly

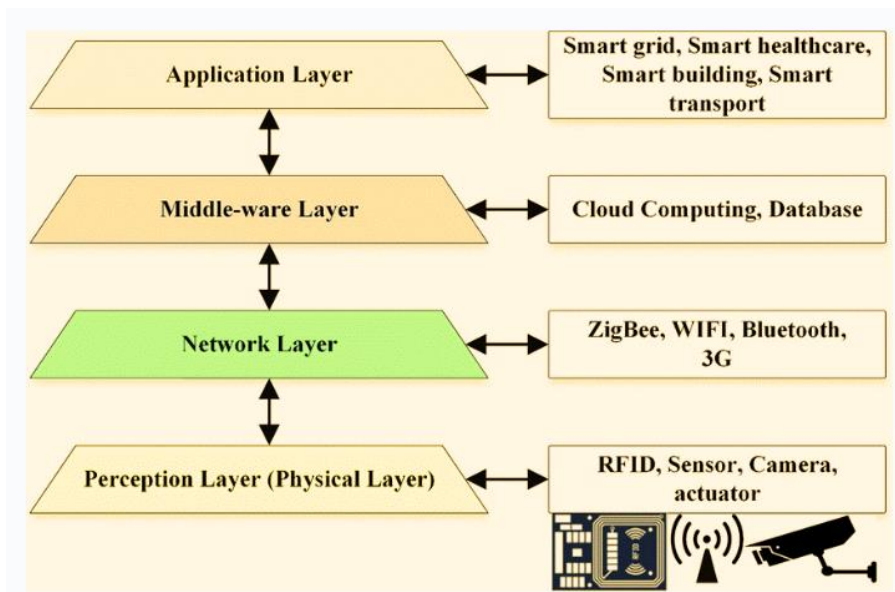


Figure 1. An organized overview of the four distinct IoT layers.

reduces the risk of unauthorized access by requiring users to verify their identity using multiple authentication factors like passwords, biometrics, or one-time codes. Additionally, attackers may be prevented from taking advantage of weak or default passwords by enforcing strong password policies and regularly updating credentials. Encryption of data is yet another crucial aspect of IoT security. By encrypting data sent between IoT devices, attackers cannot decipher the information even if they intercept the communication (Roman et al., 2013; Singh et al., (2024); Sicari et al., 2015). Strong protection against data breaches and unauthorized access is provided by advanced encryption standards such as AES-256 and RSA.

The IoT Architecture, depicted in Figure 1, consists of four primary layers: the Application Layer, the Network Layer, the Middleware Layer, and the Perception Layer (Physical Layer). In IoT systems, communication, data processing, and utilization are all made possible by each of these layers. The environment's sensing and data collection are the responsibility of the Perception Layer, also called the Physical Layer. It includes sensors, cameras, actuators, and RFID (Radio Frequency Identification) devices that detect physical changes like temperature, light and motion. Stankovic (2014) and Sharma & Park (2018) described this layer as the interface between the Internet of Things system and the real world, collecting data and sending it to the next layer for further processing. The seamless transfer of data from the Perception Layer to other IoT system components is made possible by the Network Layer. It facilitates data transfer between IoT devices, cloud servers and other data processing units by utilizing a variety of communication technologies like ZigBee, WiFi, Bluetooth, and 3G. The upkeep of secure and dependable connectivity within IoT networks depends on this layer. The Middleware Layer acts as a bridge between the Application Layer and the Network Layer. Using cloud computing and database technology, it is in charge of processing, managing, and storing data. Before being sent to applications for automation and decision-making, this layer makes sure that the raw sensor data from IoT devices is processed effectively (Tahsien et al., 2020).

The highest layer that provides IoT services to end users is the Application Layer. It incorporates processed data into a variety of real-world applications, such as smart buildings, smart transportation systems, smart healthcare, and smart grids. This layer ensures that IoT-generated insights are accessible and useful for improving automation, monitoring, and real-time decision-making. The goal of IoT architecture is to make

it easier for devices in the Internet of Things to communicate and process data in a timely manner. Beginning with data collection at the Perception Layer, data is transmitted via the Network Layer, processed at the Middleware Layer, and useful insights are provided at the Application Layer. This structured approach enhances the efficiency, scalability and security of IoT systems (Uprety and Rawat, 2021; Vasilomanolakis et al., 2015). By ensuring that data remains protected throughout its entire transmission journey—from the Internet of Things device to the cloud server or user interface- end-to-end encryption further enhances security. Machine learning (ML) and artificial intelligence (AI) are also making a big difference in making IoT security better. Real-time analysis of network traffic patterns, anomalies, and potential threats are all possible with AI-powered security solutions. Machine learning algorithms can continuously improve their threat detection capabilities and learn from previous security incidents, allowing for quicker and more accurate responses to new cyber threats. IoT networks can be monitored for suspicious activities like unauthorized login attempts, unusual data transfers, or malware infections by AI-based intrusion detection systems (IDS) and they can take immediate action to prevent security breaches. In addition, blockchain technology is gaining popularity as a promising approach to protecting IoT ecosystems. By leveraging decentralized and tamper-proof ledger systems, blockchain can provide transparent and secure communication between IoT devices, reducing the risk of data manipulation and unauthorized access. Identity management systems based on the blockchain can also improve authentication procedures, ensuring that only reputable devices can join IoT networks. Governments and regulatory bodies have a crucial role to play in strengthening IoT security (Haque and Tasmin, 2020). Global security standards and regulations can help ensure that IoT device designers and developers adhere to the best security practices. IoT ecosystems' security can be significantly improved by regulations that require data protection measures, vulnerability disclosures, and regular security updates. Proactive security measures can also be encouraged by raising consumer and business awareness of IoT security risks and best practices. The likelihood of security incidents can be decreased by educating users about the significance of changing default passwords, activating security features, and keeping devices up to date (Kumar and Lee, 2020; Okporokpo et al., 2023). IoT solution providers should also conduct routine security assessments, implement network segmentation to separate IoT devices from

essential systems, and develop incident response plans to lessen the impact of potential security breaches (Kumar et al., 2023). The sophistication of cyber threats will increase with IoT technology. Ransomware targeting IoT devices, AI-powered cyberattacks, and supply chain vulnerabilities are examples of new threats that call for ongoing IoT security research and development. Radoglou-Grammatikis et al. (2020) said that the creation of self-healing security systems that can identify and fix security flaws on their own holds great promise for strengthening the IoT's defenses against cyberattacks. In addition, new encryption techniques that can withstand future cyberattacks may be made available by developments in quantum cryptography. However, manufacturers, governments, cybersecurity experts, and end users must work together to create a secure IoT ecosystem. The Internet of Things (IoT) industry has the potential to create a connected world that is safer and more reliable by prioritizing security from the design phase, implementing proactive security measures, and staying ahead of emerging threats (Weber, 2010; Miorandi et al., 2012).

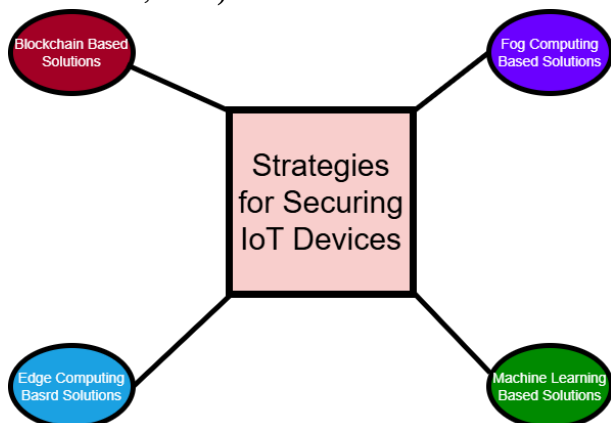


Figure 2. Key Strategies for Securing IoT Devices.

As shown in Figure 2, a combination of cutting-edge technologies is required to protect IoT devices from emerging security threats. Blockchain-based solutions, which offer a security framework that is decentralized and resistant to tampering, are one effective strategy. Blockchain safeguards data integrity and prevents unauthorized access by utilizing cryptographic hashing and smart contracts. It reduces the likelihood of identity theft and cyberattacks because there is no need for a central authority. Fog computing-based solutions, which extend cloud computing by processing data closer to IoT devices, are another crucial strategy. This improves real-time threat detection and reduces latency. By distributing workloads more effectively, fog computing also makes it possible to implement localized security policies and detect anomalies. This makes risks like Distributed Denial-of-Service (DDoS) attacks less likely. In a similar

vein, edge computing-based solutions boost security by processing data locally on Internet of Things (IoT) devices or nearby gateways (Issarny et al., 2011). Edge computing lowers the likelihood of data being intercepted by cybercriminals by limiting data exposure to external networks. In addition, it makes it possible to implement real-time security measures like authentication protocols and intrusion detection, making it particularly useful in sectors like healthcare and industrial IoT where low-latency security is essential. Finally, IoT security relies heavily on machine learning-based solutions that use AI-powered models to identify and mitigate threats. In real-time, machine learning algorithms look at network traffic, find anomalies, and predict possible security risks. These models are highly effective against evolving cyberattacks because they continuously improve by learning from new threats. IoT security can be significantly enhanced by integrating blockchain, fog computing, edge computing, and machine learning, resulting in a connected environment that is safer and more resilient (Botta et al., 2016).

To safeguard devices from emerging cyber threats, significant security issues have emerged as a result of the IoT's rapid expansion. IoT ecosystems are at risk due to a number of primary vulnerabilities, including inadequate security updates, weak authentication, encryption, and standardized security protocols. However, it is possible to enhance IoT security and mitigate potential risks by implementing strong authentication mechanisms, encryption techniques, AI-driven threat detection, and regulatory frameworks (Zhang et al., 2014). Innovative security solutions that safeguard IoT devices and ensure a secure digital future will require ongoing research and collaboration as cyber threats continue to evolve.

Materials and Methods

A comprehensive research methodology that combines qualitative and quantitative approaches is used in this study on protecting IoT devices from emerging security threats. To evaluate the efficacy of various security solutions, a literature review, threat modeling, experimental analysis, and simulation-based evaluations are used.

Encryption

A key expansion function in AES (Advanced Encryption Standard) is used to generate round keys from the initial secret key.

$$W[i] = W[i - Nk] \oplus T(W[i - 1]) \quad (1)$$

RSA Encryption and Decryption

In IoT, secure key exchanges are frequently carried out using RSA (Rivest-Shamir-Adleman).

For Ciphertext

$$C = Me \bmod N \quad (2)$$

For Original Message

$$M = Cd \bmod N \quad (3)$$

Entropy-Based Anomaly Detection for Intrusion Detection

Entropy measures the randomness of network traffic to identify anomalies.

$$H(X) = -\sum p(x) \log_2 p(x) \quad (4)$$

Blockchain Security: Hashing Function (SHA-256)

Hash functions like SHA-256 are necessary for blockchain-based IoT security.

$$H = \text{SHA} - 256(M) \quad (5)$$

Probability of Authentication Using Multi-Factor Authentication (MFA)

By requiring multiple authentication factors, MFA boosts IoT security.

$$P_{\text{auth}} = P_1 * P_2 * P_3 \quad (6)$$

#Threat Modeling Approach

Threat modeling was used to classify and investigate IoT ecosystem-specific security threats. STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) and DREAD (Damage Potential, Reproducibility, Exploitability, Affected Users, Discoverability) frameworks were utilized in the implementation of a structured strategy. This made it possible to categorize security threats and gain insight into how they might affect IoT infrastructure.

#Experimental Setup

An experimental testbed was built with smart sensors, Internet of Things (IoT) devices like the Raspberry Pi, Arduino, and others to see how well security solutions work. Distributed Denial of Service (DDoS), Man-in-the-Middle (MITM), and data exfiltration attacks were all controlled cyberattacks on the devices. Cryptographic protocols, firewalls, and intrusion detection systems are just a few of the security mechanisms that have been implemented and evaluated for their effectiveness in reducing threats.

#Simulation-Based Evaluation

NS3 and MATLAB were used to simulate huge IoT networks for a simulation-based analysis. Security measures, such as machine learning-based anomaly detection and lightweight encryption algorithms, were evaluated for their effectiveness in real-time threat detection after a variety of attack scenarios were modeled.

#Data Collection and Analysis

Data was gathered using both simulations and real-time experiments. Response time, latency, packet loss, and detection accuracy were all recorded as performance metrics. Statistical analysis was carried out to evaluate the efficacy of various security strategies and compare them.

#Ethical Considerations

To ensure ethical compliance, all experiments were carried out in a controlled environment. IoT networks in the real world were not hacked, and security measures were only evaluated for research purposes. A strict adherence to cybersecurity research best practices ensured that neither users nor systems were harmed.

#Security Policy Analysis

A review of the existing IoT security policies and regulatory frameworks was carried out in addition to the technical evaluations. The goal of this analysis was to find loopholes in the rules that are in place now and suggest changes to make IoT devices more secure and private. Effective security policies were proposed after compliance with industry standards like GDPR, NIST Cybersecurity Framework, and ISO/IEC 27001 were examined.

#Hardware and Software Used

For security analysis, the study used a mix of open-source and commercial tools. Software tools like Wireshark for network analysis, TensorFlow for AI-based threat detection, and OpenVAS for vulnerability assessment were utilized, as were microcontrollers, edge computing devices, and wireless sensors.

#User Behavior and Security Awareness

In the context of IoT security, user behavior was analyzed to complement technical solutions. The level of security awareness among IoT users was assessed through surveys and interviews. The impact of human factors, like how passwords are managed and how vulnerable people are to phishing attacks, was looked at in order to make suggestions for making users more aware of security.

A structured approach to protecting IoT devices, beginning with threat identification, is depicted in Figure 3. Identifying potential security flaws and threats that could compromise IoT systems is the first step. To ensure a proactive security strategy, cyber threats like malware, unauthorized access, and data breaches are identified. The process moves on to risk assessment and analysis after threats are identified, where their likelihood and impact are evaluated. Organizations can use this step to effectively prioritize security measures and allocate resources. Encryption, authentication protocols, and

firewall configurations are examples of protective measures that are incorporated into security implementation following risk assessment. To safeguard data transmission, device communication, and system access, security policies are enforced. Real-time threat detection plays a crucial role in identifying potential attacks as they occur after implementation. Artificial intelligence (AI)-powered solutions, anomaly detection algorithms, and intrusion detection systems all aid in the immediate identification and mitigation of security threats. It is necessary to maintain constant monitoring and updates to guarantee ongoing protection. To keep up with changing cyber risks, security systems must be updated frequently with patches, software upgrades, and threat intelligence. Monitoring the behavior of devices and the network on a regular basis aids in the detection of unusual patterns and the prevention of security breaches. Last but not least, compliance and standardization guarantee that IoT security measures adhere to global cybersecurity standards and regulations. User and stakeholder trust is increased and overall security is improved by adhering to industry standards and legal requirements. IoT device protection is improved and risks are reduced thanks to this structured approach, which ensures a comprehensive security framework.

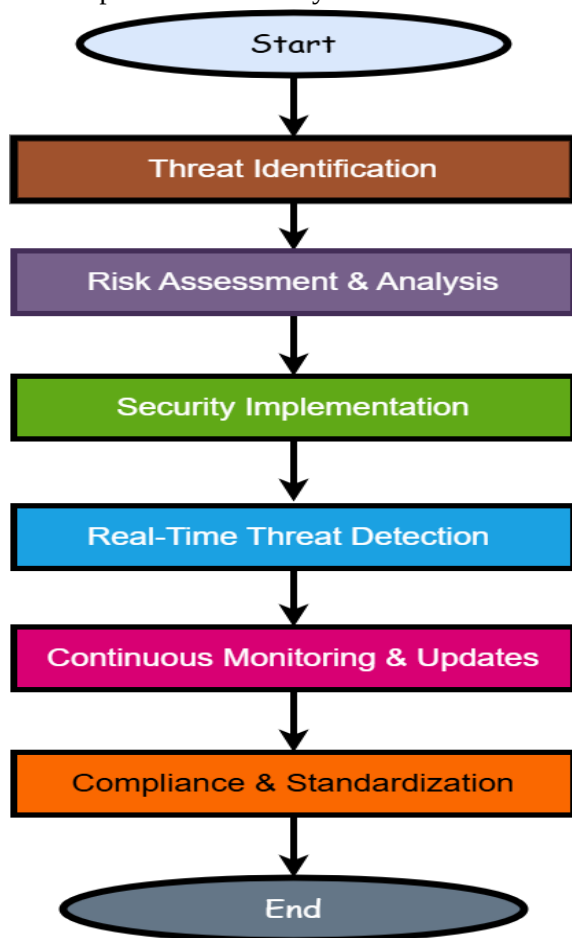


Figure 3. Workflow for Threat Detection and Protection in the IoT Security Framework.

Combining these approaches makes a thorough and trustworthy evaluation of IoT security issues and solutions possible, aiding in the creation of solid security frameworks for future IoT implementations.

Result and Discussion

Due to the increasing complexity of interconnected devices, the analysis of IoT security threats and countermeasures reveals that emerging risks continue to evolve.

#Encryption Performance Comparison

Table 1 compares encryption algorithms in terms of execution time and energy consumption. Blowfish is the fastest and most efficient, followed by AES and ECC, while RSA is the slowest and uses the most energy.

The execution time (ms) and energy consumption (J) of the various encryption algorithms (AES, RSA, ECC, and Blowfish) are compared in Figure 4. RSA has the longest execution time on the execution time curve, indicating that it is computationally demanding. In contrast, Blowfish and ECC have shorter execution times, making them more effective. RSA consumes the most energy, while Blowfish and AES consume the least, as evidenced by the energy consumption curve. Based on efficiency and resource consumption, this analysis aids in selecting the best encryption algorithm for IoT and security applications.

Table1. Comparison of encryption algorithm performance.

Encryption Algorithm	Execution Time (ms)	Energy Consumption (J)
AES	12.5	3.2
RSA	45.8	8.9
ECC	18.3	5.1
Blowfish	9.7	2.8

The findings emphasize that, despite significant progress in IoT security, challenges remain due to the dynamic nature of threats, the limited resources of IoT devices, and the requirement for scalable, real-time protection mechanisms.

***Balancing Performance and Security:** Implementing robust security measures without sacrificing device performance remains one of the greatest obstacles. Although lightweight encryption protocols like elliptic curve cryptography (ECC) have been proposed, it is necessary to take into account tradeoffs between security and computational efficiency.

***Problems with interoperability and standardization:** Because IoT devices and manufacturers are so different, there aren't many standard security protocols. This fragmentation leaves gaps that attackers can exploit and prevents widespread adoption of universal security frameworks.

***Threat Mitigation Driven by AI:** Although AI-based security systems improve threat detection, they are susceptible to adversarial learning, in which malicious individuals manipulate models by injecting false data. The creation of resilient AI models is still a major area of research focus.

***Challenges in Regulatory and Compliance:** Despite the introduction of IoT security regulations by governments and regulatory bodies, compliance remains inconsistent. A more secure IoT ecosystem can be created with the adoption of best practices like Zero Trust Architecture (ZTA) and Secure Boot Mechanisms.

***Future Research Directions:** To address upcoming security issues, quantum-safe cryptography, federated learning for secure AI training, and self-healing IoT systems are crucial. To coincide with the low-power nature of IoT devices, research must also concentrate on energy-efficient security solutions.

0.05 and the TPR is 0.82. There is a trade-off between catching more attacks and generating more false alarms as the FPR rises, as does the TPR. At FPR = 0.25, the TPR reaches 0.96, meaning 96% of anomalies are detected, but 25% of normal behaviors are falsely identified as attacks.

Table 2. Ratio of False Positives to Rate of True Positives for Finding Anomalies.

False Positive Rate (FPR)	True Positive Rate (TPR)
0.05	0.82
0.10	0.89
0.15	0.91
0.20	0.94
0.25	0.96

***Graphical Representation**

ROC Curve

The ROC curve plots TPR vs. FPR, showing the

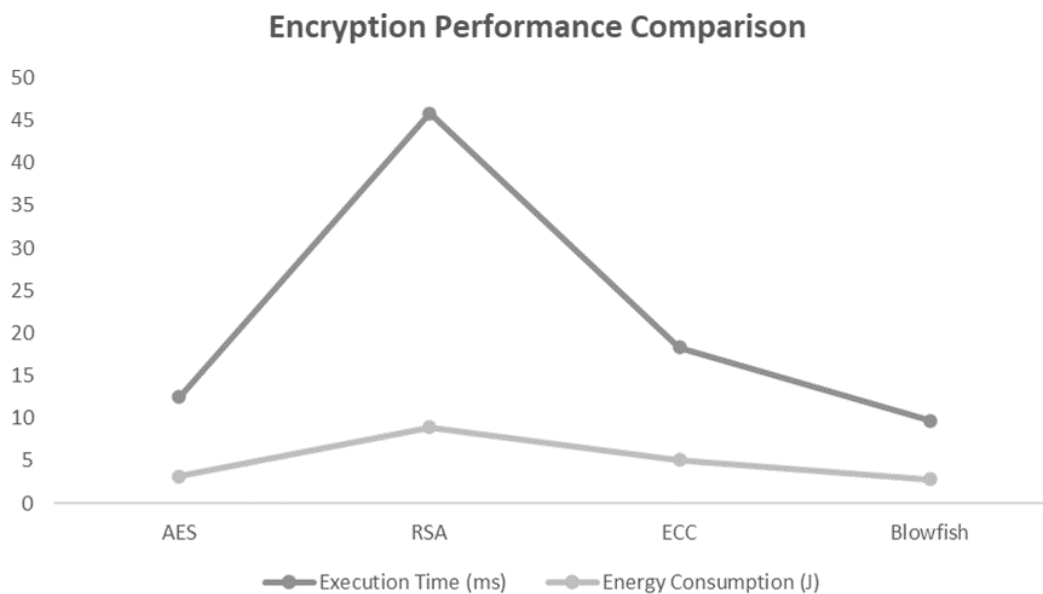


Figure 4. Encryption Algorithm Performance Comparison.

#Anomaly Detection Accuracy

In IoT security, anomaly detection is crucial for identifying malicious activities or unauthorized access. The Receiver Operating Characteristic (ROC) Curve is a popular metric for assessing an anomaly detection system's effectiveness. The True Positive Rate (TPR) is plotted against the False Positive Rate (FPR) to assess a system's ability to differentiate between legitimate and malicious behavior.

***Data Interpretation**

The relationship between an IoT security anomaly detection system's True Positive Rate (TPR) and False Positive Rate (FPR) is shown in Table 2.

***Analysis of the Data**

The system correctly detects 82% of anomalies and only falsely flags 5% of normal events when the FPR is

trade-off between sensitivity and specificity. A well-performing system has a curve that is closer to the top-left corner (high TPR, low FPR). The Area Under the Curve (AUC) indicates the overall effectiveness of the model. A value close to 1 suggests a strong anomaly detection model.

Line Graph Representation

A line graph can illustrate how TPR improves with increasing FPR. A steeper initial rise (low FPR, high TPR) indicates an effective system. The model might not be able to effectively differentiate between legitimate and malicious actions if the graph is nearly linear. To prevent false alarms, the anomaly detection model used in IoT security should strive for a high TPR while maintaining a low FPR. Using machine learning or AI-driven methods, a well-trained system can strike a balance between

accurately identifying threats and minimizing devices from emerging threats. It is possible to create a safer Internet of Things ecosystem by implementing unnecessary alerts.

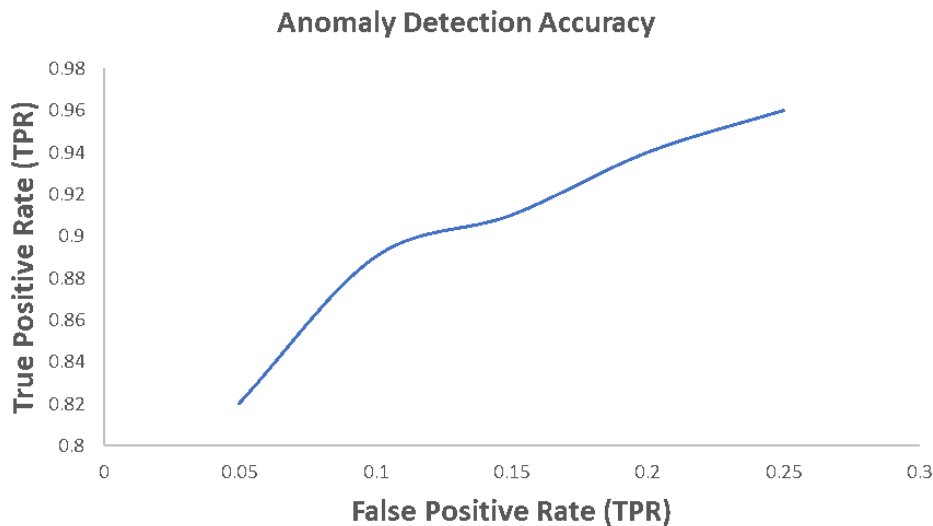


Figure 5. Anomaly Detection Accuracy ROC Curve Representation.

The trade-off between the true positive rate (TPR) and the false positive rate (FPR) for anomaly detection is depicted in a ROC curve in Figure 5. The findings suggest that a multi-layered security strategy that incorporates lightweight cryptography, edge computing security, AI-driven threat detection, blockchain-based authentication, and AI-driven threat detection can significantly improve IoT protection. In order to guarantee the long-term safety of IoT networks, future research ought to focus on enhancing AI-based security systems' interoperability, regulatory compliance, and adversarial resilience.

Conclusion

Connectivity and productivity in a variety of industries have been significantly enhanced as a result of the rapid adoption of IoT technology. However, serious threats to the security of IoT devices pose serious threats to data privacy, device integrity, and system functionality as a whole. The absence of standardized security protocols, weak authentication mechanisms, and unencrypted data transmission are just a few of the key security issues highlighted in this paper. Strong authentication, encryption methods, regular security updates, and the integration of AI and ML-based threat detection systems are all needed to address these vulnerabilities. To further reduce IoT security risks, governments, businesses, and manufacturers need to enforce standard security regulations and best practices. IoT security can be further strengthened by raising user awareness of cybersecurity threats and encouraging responsible device management. Continuous innovation, teamwork, and vigilance are necessary to protect IoT

robust security measures and maintaining proactive threat mitigation strategies. In order to ensure the long-term protection of IoT infrastructure, future research should concentrate on developing adaptive security solutions that can adapt to new and sophisticated cyber threats.

Conflict of interest

None

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How to cite this Article:

Naman Tiwari, Abhay Kumar Pandey and Abhilash Kokala (2025). Challenges and Solutions for Countering Emerging Security Threats to IoT Devices. *International Journal of Experimental Research and Review*, 47, 59-68.

DOI : https://doi.org/10.52756/ijerr.2025.v47.005



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